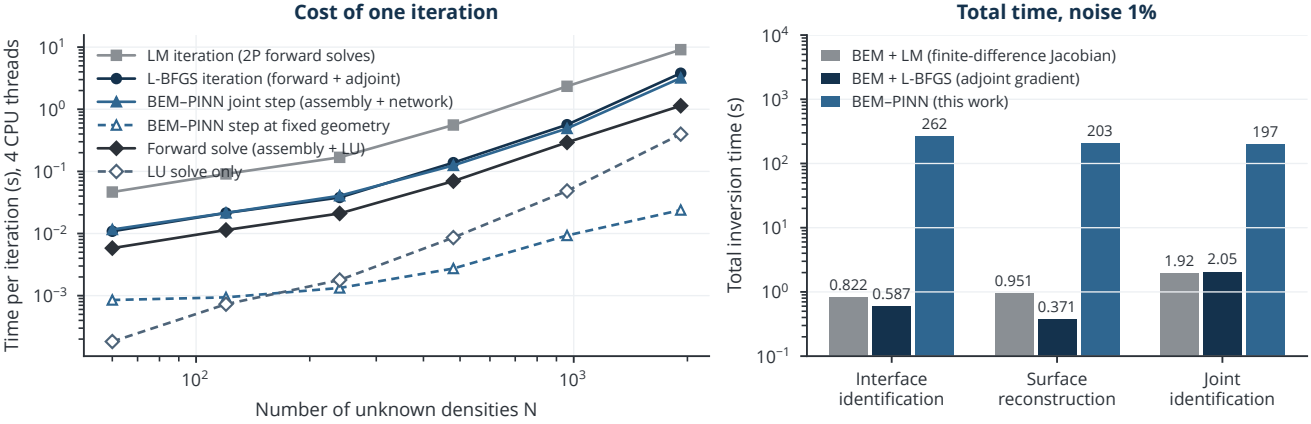
Cost of one iteration
LM iteration (2P forward solves)
L-BFGS iteration (forward + adjoint)
BEM–PINN joint step (assembly + network)
BEM–PINN step at fixed geometry
Forward solve (assembly + LU)
LU solve only
Time per iteration (s), 4 CPU threads
Number of unknown densities N
Total time, noise 1%
BEM + LM (finite-difference Jacobian)
BEM + L-BFGS (adjoint gradient)
BEM–PINN (this work)
Total inversion time (s)
0.822
0.587
262
0.951
0.371
203
1.92
2.05
197
Interface identification
Surface reconstruction
Joint identification